If the car turns with the speed little greater than, then the static friction acts down the slope to prevent outward skidding. But if the car turns with the speed much greater than v_{safe} then static friction cannot prevent from outward skidding.

SOLVED EXAMPLE UNIT-4

1. A force $\vec{F} = \hat{i} + 2\hat{j} + 3\hat{k}$ N acts on a particle and displaces it through a distance $\vec{S} = 4\hat{i} + 6\hat{j}$ m. Calculate the work done.

Solution:

Force $\vec{F} = \hat{i} + 2\hat{j} + 3\hat{k}$ N

Distance $\vec{S} = 4\hat{i} + 6\hat{j}$ m

Work done = $\vec{F} \cdot \vec{S} = (\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (4\hat{i} + 6\hat{j})$

= 4+12+0 = 16 J

2. A particle moves along X- axis from x=0 to x=8 under the influence of a force given by F= $3x^2 - 4x + 5$. Find the work done in the process.

Solution:

Work done in moving a particle from x=0 to x=8 will be

$$W = \int_0^8 F dx = \int_0^8 (3x^2 - 4x + 5) dx$$

$$= \left[\frac{3x^3}{3} - \frac{4x^2}{2} + 5x \right]_0^8$$

$$W = \left[3\frac{(8)^3}{3} - 4\left(\frac{8^2}{2}\right) + 40 \right]$$

$$= [512 - 128 + 40] = 424 J$$

3. A body of mass 10kg at rest is subjected to a force of 16N. Find the kinetic energy at the end of 10 s.

Solution:

Mass m = 10 kg

Force F = 16 N

time t = 10 s

$$a = F/m = \frac{16}{10} = 1.6 m s^{-2}$$

we know that, $v = u + at$

$$= 0 + 1.6 \times 10 = 16 \, m s^{-1}$$

Kinetic energy K.E = $\frac{1}{2} mv^2$

$$= \frac{1}{2} \times 10 \times 16 \times 16$$

$$= 1280 J$$

4. A body of mass 5kg is thrown up vertically with a kinetic energy of 1000 J. If acceleration due to gravity is 10 $m s^{-2}$, find the height at which the kinetic energy becomes half of the original value.

Solution:

Mass m = 5 kg

K.E E = 1000 J

g = 10 $m s^{-2}$

At a height 'h', mgh = $\frac{E}{2}$

$$5 \times 10 \times h = \frac{1000}{2}$$

$$h = \frac{500}{50} = 10 m$$

5. Two bodies of masses 60 kg and 30 kg moving in the same direction along straight line with velocity 40 cm s^{-1} and 30 cm s^{-1}

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